

TRAINING-FREE CONTEXTUAL ASR VIA SPEECHLLM-BASED ERROR-AWARE SELECTIVE RETRIEVAL

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ABSTRACT

Recognition of domain-specific and low-frequency terms remains challenging for automatic speech recognition (ASR). Although contextual biasing can improve their recognition, directly providing a large terminology dictionary introduces many irrelevant biasing terms. Retrieval-based contextual biasing addresses this issue by selecting candidate terms from an external dictionary, but querying many recognized words requires numerous dictionary lookups and may yield poorly targeted candidates. We propose a training-free contextual ASR framework in which a pretrained speech large language model (SpeechLLM) jointly generates an ASR hypothesis and localizes error spans likely to involve domain-specific terms. Only the localized spans are used to retrieve phonologically similar terms from an external terminology dictionary. The same SpeechLLM then re-recognizes the audio conditioned on the first-pass hypothesis and the retrieved terms, without task-specific model training. To assess applicability across domains, we evaluate the framework on medical, air traffic control, and financial speech. The proposed method substantially reduces dictionary queries while improving the recall and ranking of relevant terminology candidates and second-pass ASR performance across all three domains.

Index Terms— automatic speech recognition, speech large language models, contextual biasing

1. INTRODUCTION

Recognition of domain-specific and low-frequency terms remains challenging for automatic speech recognition (ASR) [1, 2]. Contextual biasing can improve the recognition of such terms by providing candidate words or phrases at inference time [1, 3, 4]. Recent speech large language model (SpeechLLM)-based approaches have further incorporated pronunciation-aware context and bias-word information [5, 6, 7]. However, directly using a large terminology dictionary introduces many irrelevant candidates, increasing computational cost and potentially causing over-biasing [2, 7, 8].

To address this issue, prior work has explored several strategies for selecting terms or regions for retrieval or biasing, including named-entity-based retrieval [9], phonetic retrieval [10], learned retrieval [2, 11], bias-word ranking and selection [12], terminology relevance estimation [13], and localized alignment [14]. Several recent approaches rely on separately trained retrieval, ranking, or alignment components [2, 11, 12, 14], introducing additional training and model-development requirements. Moreover, existing selection criteria do not explicitly target regions that are likely to correspond to domain terms and contain recognition errors. This motivates a training-free approach to determining whether terminology retrieval is needed and which regions should trigger it.

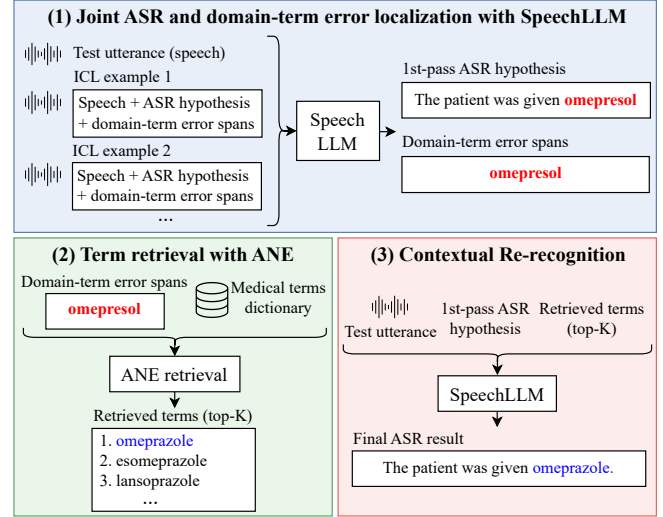


Fig. 1: Overview of the proposed training-free contextual ASR framework with error-aware selective terminology retrieval.

In parallel, ASR error detection has been studied to guide selective correction and decoding [15, 16, 17]. However, detected errors have primarily been used to determine where correction or re-decoding should be applied, rather than whether an external terminology dictionary should be queried.

SpeechLLMs provide a natural basis for this problem because they can jointly exploit acoustic information and textual context while generating an ASR hypothesis [18]. If the same pretrained model can also identify domain-term error regions in its own hypothesis, terminology retrieval can be triggered without training a separate error-detection model, thereby avoiding additional task-specific model development. This motivates the following question: can a pretrained SpeechLLM identify domain-term recognition errors in its own ASR hypothesis and use them to determine whether terminology retrieval is needed and which regions should trigger it?

To address this question, we propose a training-free contextual ASR framework in which a pretrained SpeechLLM jointly generates an ASR hypothesis and localizes error spans likely to involve domain-specific terms in a single inference. Rather than directly correcting the localized errors, we use the predicted spans as a hard gate for terminology retrieval, such that only regions likely to contain domain-term recognition errors trigger dictionary search. Localized spans are used to retrieve phonologically similar domain terms using Acoustic Neighbor Embeddings (ANE) [19]. The same SpeechLLM then re-recognizes the audio conditioned on the first-pass ASR hypothesis and the retrieved terms. The entire framework requires no

task-specific model training.

Experiments on medical, air traffic control, and financial speech show that the proposed method substantially reduces dictionary queries while improving the recall and ranking of relevant terminology candidates and second-pass ASR performance.

2. PROPOSED METHOD

We propose a training-free contextual ASR framework in which SpeechLLM-based domain-term error localization is used to selectively trigger retrieval from an external terminology dictionary. As illustrated in Fig. 1, the framework consists of three steps: (1) joint generation of a first-pass ASR hypothesis and localization of error spans likely to involve domain-specific terms, (2) terminology retrieval using only the localized spans as queries, and (3) re-recognition of the same audio conditioned on the first-pass ASR hypothesis and the retrieved terms. The localized spans serve as a hard gate for dictionary retrieval, such that only regions predicted to contain domain-term recognition errors trigger terminology search. The same pretrained SpeechLLM is used for joint first-pass ASR and domain-term error localization and for second-pass contextual re-recognition, with no task-specific parameter updates.

2.1. Joint ASR and Domain-Term Error Localization

Let $\mathbf{x}$ denote an input speech signal, and let p_{dom} denote the dataset-specific localization guidance provided in the prompt. As exemplified for MedSyn in Table 1, the guidance specifies the target domain, relevant terminology categories, and selection criteria requiring a span to be both domain-relevant and likely to contain an ASR error. The same task, selection criteria, ICL format, and output format are used across datasets, while the domain description, terminology categories, and demonstrations are adapted to each domain.

Given $\mathbf{x}$ and p_{dom} , the SpeechLLM jointly generates a first-pass ASR hypothesis $\mathbf{h}^{1\text{st}}$ and localizes a set of domain-term recognition error spans $\mathcal{S} = \{s_1, \dots, s_M\}$. Each s_m is a short text span extracted from $\mathbf{h}^{1\text{st}}$ and predicted to contain a recognition error involving a domain-specific term. After generation, the predicted spans are normalized and restricted to at most two words, and numeric-only spans are discarded.

We perform joint ASR and domain-term error localization using audio-text in-context learning (ICL) [20]. Let

$$\mathcal{D}_{\text{ICL}} = \{(\mathbf{x}_i, \mathbf{h}_i, \mathcal{S}_i)\}_{i=1}^J \quad (1)$$

denote the set of J ICL demonstrations, where $\mathbf{x}_i$ is an example speech signal, $\mathbf{h}_i$ is its ASR hypothesis, and $\mathcal{S}_i = \{s_{i,1}, \dots, s_{i,M_i}\}$ is the corresponding set of domain-term recognition error spans. We design the ICL demonstrations to cover four recognition cases: an error-free case, a domain-term substitution, a domain term misrecognized as a common-word sequence, and a phonetically similar substitution. Each demonstration follows the same structured output format as the test inference, containing an ASR hypothesis followed by its domain-term error spans. For each error-containing demonstration, only the ASR-side span corresponding to the misrecognized domain term is annotated in $\mathcal{S}_i$, whereas the error-free demonstration uses $\mathcal{S}_i = \emptyset$. The demonstrations therefore illustrate both whether a span should be selected and where it should be localized, without providing a corrected transcript as the localization target.

For a test utterance, the structured output is serialized such that the complete first-pass ASR hypothesis is generated before the domain-term error spans. Since autoregressive generation conditions later output tokens on preceding generated tokens [18, 21],

Table 1: Prompt example for joint ASR and domain-term error localization on MedSyn.

Task: Transcribe the input speech and identify short domain-term error spans in the generated transcript.
Domain guidance: The target domain is medical speech. Relevant terminology includes medications, diseases, anatomy, clinical procedures, laboratory tests, and medical abbreviations.
Selection guidance: Select a span only when it is both domain-relevant and likely to be incorrect. Do not select a span solely because it contains a domain-specific term. Use acoustic evidence and the generated transcript to identify misrecognized, weakly supported, or acoustically unclear regions. Return short spans corresponding to the ASR-side error expression.
ICL format: Each demonstration consists of an input speech signal and a structured output containing its ASR hypothesis and domain-term error spans. The demonstrations include both error-containing and error-free cases.
Output: {"transcript": "...", "error-spans": [...]}

this ordering allows span localization to condition directly on the SpeechLLM’s own generated hypothesis. The joint inference is expressed as

$$(\mathbf{h}^{1\text{st}}, \mathcal{S}) = F_{\text{SLLM}}(\mathbf{x}, p_{\text{dom}}; \mathcal{D}_{\text{ICL}}). \quad (2)$$

Unlike post-hoc localization applied to a fixed ASR hypothesis, the proposed method jointly obtains the first-pass hypothesis and its domain-term recognition error spans within a single SpeechLLM inference. The localized spans are not corrected directly at this stage. Instead, they determine whether terminology retrieval is needed and which regions should trigger it. Only the predicted spans are used as retrieval queries, while the complete first-pass hypothesis is retained for subsequent second-pass re-recognition but is not used to query the terminology dictionary.

2.2. Terminology Retrieval from Localized Spans

Each localized span $s_m \in \mathcal{S}$ is used as an independent retrieval query, yielding $\mathcal{Q} = \{s_1, \dots, s_M\}$. Let $\mathcal{V} = \{v_1, \dots, v_L\}$ denote an external terminology dictionary containing L domain-specific terms. We retrieve terms that are phonetically similar to the localized spans using Acoustic Neighbor Embeddings (ANE) [19]. Let $g(\cdot)$ denote the ANE text encoder. For a query q and dictionary term v , the similarity score is computed as

$$r(q, v) = \frac{1}{1 + \|g(q) - g(v)\|_2}. \quad (3)$$

When multiple spans are localized, the similarity scores are aggregated for each dictionary term as

$$R(v \mid \mathcal{Q}) = \max_{q \in \mathcal{Q}} r(q, v). \quad (4)$$

The dictionary terms are ranked by $R(v \mid \mathcal{Q})$ across all localized spans, and the top- K terms form the candidate set $\mathcal{C}_K$. If no span is localized ($M = 0$), terminology retrieval and second-pass re-recognition are skipped, and the first-pass hypothesis is retained.

2.3. Second-Pass Contextual Re-recognition

The retrieved terms $\mathcal{C}_K$ and the first-pass ASR hypothesis $\mathbf{h}^{1\text{st}}$ are provided to the same SpeechLLM together with the input speech for second-pass re-recognition. The retrieved terms serve as contextual biasing information [1, 3], while the first-pass hypothesis provides textual context for re-recognition. The resulting second-pass ASR hypothesis is expressed as

$$\mathbf{h}^{2\text{nd}} = F_{\text{SLLM}}(\mathbf{x}, \mathbf{h}^{1\text{st}}, \mathcal{C}_K). \quad (5)$$

Table 2: Statistics of the evaluation datasets.

Dataset	Number of utterances	Reference words	Domain terms	Domain ratio (%)
ATCOSIM [22]	1,901	22,642	167	9.00
Earnings [23, 24]	772	26,750	652	5.04
MedSyn [25]	7,906	106,566	7,198	11.65

Table 4: Domain-term error localization performance on MedSyn. All metrics except Spans/Utt. are reported in percent.

Method	Spans /Utt.	Error Precision	Error Recall	General Recall	Domain Recall
Span-selection baselines					
All ASR 1-grams	13.37	8.1	92.8	83.6	99.1
Random span	0.99	11.5	10.6	11.0	10.3
NE extraction	1.72	29.5	63.6	37.0	81.7
Confidence span	1.15	49.4	50.6	28.9	65.3
SpeechLLM input configurations					
ASR only	0.67	65.0	54.4	36.6	66.4
Audio only	0.94	27.4	18.6	12.3	22.8
Audio + ASR	1.07	49.9	67.0	41.4	84.4
Joint zero-shot	1.09	46.7	75.6	49.7	85.9
Ours (Joint ICL)	0.99	51.9	76.1	47.6	86.0

3. EXPERIMENTS

3.1. Experimental Setup

We evaluate the proposed method on three domain-specific speech datasets: United-MedSyn [25] (MedSyn) for medical speech, ATCOSIM [22] for air traffic control, and Contextual Earnings-22 [23, 24] (Earnings) for financial speech. Owing to the large size of MedSyn, we randomly sample 7,906 utterances, corresponding to 10% of the dataset, for all experiments. Table 2 summarizes the evaluation sets and terminology dictionaries, where the domain ratio denotes the proportion of reference words corresponding to dictionary terms. Following prior work [26, 27], we construct each terminology dictionary from reference transcriptions by retaining target-domain unigrams occurring at most once per million words in LibriSpeech [28], Common Voice [29], and GigaSpeech [30]. Number words, words shorter than three characters, and non-alphabetic tokens are excluded. The resulting dictionary is fixed across evaluation utterances, and utterance-level reference information is not used during retrieval.

We use Qwen3-Omni-30B-A3B-Instruct [18] as the SpeechLLM for both joint localization and second-pass re-recognition, without parameter updates. *Joint ICL* uses four audio-text ICL demonstrations, whereas all other localization methods use no demonstrations. *Joint zero-shot* uses the same localization instructions as *Joint ICL* without demonstrations. Table 3 shows the domain-term portions of the examples. The demonstrations are selected from the training set and are disjoint from the evaluation utterances. The reference-term column is shown only for interpretation and is not provided to the model. Predicted spans are limited to two words, and numeric-only spans are excluded from retrieval.

We use the publicly released pretrained ANE model `embedder-64` [31] without additional training to rank dictionary terms for each predicted span. We refer to ASR-only hypotheses generated by the same SpeechLLM as *Baseline ASR*. All non-joint methods share these fixed hypotheses, whereas *Joint zero-shot* and *Joint ICL* jointly generate their own hypotheses and spans. Retrieval and second-pass settings are otherwise shared across methods.

Table 3: Audio-text ICL examples used for MedSyn evaluation.

Error type	Reference term	ASR hypothesis	Error span
Error-free	Amoxicillin	Amoxicillin	<i>None</i>
Domain-term substitution	Hydralazine	Hydroxyzine	Hydroxyzine
Common-word substitution	Metformin	met foreman	met foreman
Phonetic substitution	ileum	ilium	ilium

3.2. Domain-Term Error Localization

We compare *Joint ICL* with four span-selection baselines, three non-joint SpeechLLM input configurations, and a joint zero-shot ablation. *All ASR 1-grams* selects every non-numeric word in the *Baseline ASR* hypothesis. *Random span* randomly samples spans while matching the number and word lengths of the *Joint ICL* spans for each utterance. *NE extraction*, motivated by named-entity-based query generation [9], prompts the same SpeechLLM to extract named entities from the *Baseline ASR* hypothesis without explicitly estimating recognition errors. *Confidence span*, motivated by conventional ASR confidence estimation and confidence-aware contextual biasing [32, 33], computes each word’s confidence as the mean log probability of its constituent decoder tokens and selects words below a dataset-specific threshold. The threshold is tuned on 500 held-out training utterances for each dataset, disjoint from the evaluation set, to maximize word-level ASR error detection F1. These span-selection methods are also used as query sources in the subsequent retrieval and contextual ASR experiments.

For the SpeechLLM input configurations, *ASR only* estimates domain-term error spans from the *Baseline ASR* hypothesis, *Audio only* estimates them from speech, and *Audio + ASR* uses both speech and the *Baseline ASR* hypothesis. *Joint zero-shot* jointly generates the first-pass ASR hypothesis and domain-term error spans within a single SpeechLLM inference without demonstrations, while *Joint ICL* performs the same joint generation using audio-text ICL.

Localization is evaluated by the average number of spans per utterance, Error Precision, Error Recall, General Recall, and Domain Recall. Recognition errors are identified by minimum-edit-distance alignment [34] and mapped to hypothesis-side positions. Deletion errors are mapped to the nearest preceding and following aligned hypothesis positions when available. Error Precision is the proportion of hypothesis-side positions covered by the predicted spans that correspond to recognition errors, whereas Error Recall is the proportion of recognition-error positions covered by the predicted spans. General Recall and Domain Recall evaluate localization of errors unrelated to and involving dictionary terms, respectively. For Domain Recall, a domain-term error is considered localized when a span overlaps its hypothesis-side error location.

Table 4 shows the localization results. *All ASR 1-grams* achieved a Domain Recall of 99.1%, but selected 13.37 spans per utterance with an Error Precision of only 8.1%, providing broad coverage at the cost of poor selectivity. In contrast, *Joint ICL* selected only 0.99 spans per utterance while achieving 51.9% Error Precision, 76.1% Error Recall, and 86.0% Domain Recall. Notably, its Domain Recall was substantially higher than its General Recall of 47.6%, indicating that the localized spans were more sensitive to recognition errors involving domain-specific terms. This may partly reflect that some ASR errors remain linguistically plausible in context, making them difficult to detect from textual cues alone [35]. Compared with *NE extraction*, *Joint ICL* reduced the number of selected spans from 1.72 to 0.99 while improving Error Precision from 29.5% to 51.9% and Domain Recall from 81.7% to 86.0%. Compared with *Confidence span*, it used fewer spans while improving Error Recall from 50.6%

Table 5: ANE retrieval performance on MedSyn. All metrics except Queries / Utt. are reported in percent.

Method	Queries					
	/Utt.	Reduction	R@1	R@10	R@50	MRR@10
All ASR 1-grams	13.37	0.0	20.7	53.2	63.7	31.9
Random span	0.99	92.6	3.7	6.1	8.4	4.4
NE extraction	1.72	87.1	30.9	52.6	64.4	38.1
Confidence span	1.15	91.4	31.3	45.4	54.3	35.8
Ours (Joint ICL)	0.99	92.6	41.4	61.8	73.1	47.9

Table 6: Second-pass ASR performance on MedSyn (%).

Method	Top-10			Top-50		
	WER	B-WER	U-WER	WER	B-WER	U-WER
Baseline ASR	7.30	41.43	2.80	7.30	41.43	2.80
All ASR 1-grams	5.83	28.83	2.80	5.99	30.42	2.77
Random span	7.12	39.95	2.79	7.09	39.81	2.78
NE extraction	5.89	29.71	2.75	6.01	30.80	2.74
Confidence span	6.08	31.40	2.75	6.22	32.52	2.75
Ours (Joint ICL)	5.67	28.05	2.72	5.88	29.79	2.72

to 76.1% and Domain Recall from 65.3% to 86.0%.

The ablation results further show that both acoustic and textual information contribute to localization. Domain Recall was 22.8% with *Audio only*, 66.4% with *ASR only*, and 84.4% with *Audio + ASR*, indicating that the two modalities provide complementary cues. *Joint zero-shot* further achieved 75.6% Error Recall and 85.9% Domain Recall. Adding audio-text ICL increased Error Precision from 46.7% to 51.9% while maintaining similar Error Recall (75.6% vs. 76.1%) and Domain Recall (85.9% vs. 86.0%), with fewer selected spans (1.09 vs. 0.99 per utterance). In addition, *Joint zero-shot* and *Joint ICL* achieved first-pass WERs of 7.59% and 7.30%, respectively, compared with 7.30% for *Baseline ASR*. Thus, *Joint ICL* localizes error spans without degrading transcription accuracy on MedSyn.

3.3. Terminology Retrieval and Contextual ASR

We first evaluate the effectiveness of the localized spans as terminology retrieval queries. Retrieval performance is measured by the average number of queries per utterance, query reduction relative to *All ASR 1-grams*, $R@K$, and $MRR@10$. $R@K$ denotes recall of domain terms involved in errors in each method’s first-pass hypothesis within the top- K candidates, while $MRR@10$ is their mean reciprocal rank, with misses in the top 10 assigned zero.

Table 5 shows the ANE retrieval results. *Joint ICL* reduced the number of queries from 13.37 to 0.99 per utterance, corresponding to a 92.6% reduction relative to *All ASR 1-grams*, while improving $R@10$ from 53.2% to 61.8% and $MRR@10$ from 31.9% to 47.9%. In contrast, *Random span*, which used the same number of queries as *Joint ICL*, achieved only 6.1% $R@10$. *Joint ICL* also achieved higher $R@K$ and $MRR@10$ than *NE extraction* and *Confidence span*.

We next evaluate second-pass ASR using the retrieved terms together with the first-pass ASR hypothesis. Following prior work on contextual ASR [27], we report overall WER, B-WER for dictionary terms, and U-WER for non-dictionary terms using the top-10 and top-50 retrieved candidates. Table 6 shows the second-pass ASR results. Compared with *Baseline ASR*, *Joint ICL* with the top-10 candidates reduced WER from 7.30% to 5.67% and B-WER from 41.43% to 28.05%, while reducing U-WER from 2.80% to 2.72%.

Table 7: Cross-domain ANE retrieval and second-pass ASR performance with top-10 retrieved candidates. All metrics except Queries / Utt. are reported in percent.

Dataset	Method	Queries					
		/Utt.	R@10	MRR@10	WER	B-WER	
ATCOSIM	Baseline ASR				14.81	46.40	
	All ASR 1-grams	5.89	66.6	34.5	13.13	32.42	
	Random span	1.08	28.1	17.1	14.08	41.00	
	NE extraction	1.51	61.5	37.5	13.06	31.97	
	Confidence span	0.97	34.3	22.9	13.68	36.93	
	Ours (Joint ICL)	1.08	69.6	46.1	12.88	30.65	
Earnings	Baseline ASR				17.97	50.93	
	All ASR 1-grams	32.71	31.8	14.8	17.68	47.15	
	Random span	1.38	5.6	3.6	17.80	49.74	
	NE extraction	2.26	35.6	22.2	17.55	46.32	
	Confidence span	1.22	17.6	12.3	17.71	46.85	
	Ours (Joint ICL)	1.38	35.8	22.5	17.45	45.66	

Despite using substantially fewer retrieval queries, *Joint ICL* also achieved lower WER and B-WER than *All ASR 1-grams* (5.67% vs. 5.83% and 28.05% vs. 28.83%, respectively). *Random span* performed considerably worse despite using the same number of queries as *Joint ICL*. For *Joint ICL*, the top-10 condition outperformed the top-50 condition, suggesting that additional lower-ranked candidates can introduce less relevant biasing terms.

3.4. Cross-Domain Validation

We evaluate the proposed method on ATCOSIM and Earnings to examine its effectiveness beyond the medical domain. Table 7 reports terminology retrieval and second-pass ASR performance. Across both datasets, *Joint ICL* substantially reduced dictionary queries while achieving the highest $R@10$ and $MRR@10$ among the evaluated query-selection methods. On Earnings, queries decreased from 32.71 to 1.38 per utterance while $R@10$ increased from 31.8% to 35.8%. *Random span*, which used a comparable number of queries, performed substantially worse than *Joint ICL*.

Joint ICL achieved first-pass WERs of 14.91% and 17.97% on ATCOSIM and Earnings, compared with 14.81% and 17.97% for *Baseline ASR*. Together with the MedSyn result, this indicates that joint localization does not substantially degrade first-pass transcription accuracy across domains.

For second-pass ASR, *Joint ICL* achieved the lowest WER and B-WER among the evaluated methods on both datasets. Compared with *Baseline ASR*, B-WER decreased from 46.40% to 30.65% on ATCOSIM and from 50.93% to 45.66% on Earnings. The smaller overall WER gain on Earnings may reflect its lower domain-term ratio. These results show that the proposed selective retrieval strategy is effective across medical, air traffic control, and financial speech.

4. CONCLUSION

We proposed a training-free contextual ASR framework using SpeechLLM-based domain-term error localization to selectively trigger terminology retrieval. The same SpeechLLM performs second-pass re-recognition using the first-pass ASR hypothesis and retrieved terms. Cross-domain experiments showed that the proposed method substantially reduces dictionary queries while improving the recall and ranking of relevant terminology candidates and second-pass ASR performance.

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